



WRO 2026 · FUTURE INNOVATORS · SENIOR

AquaBot

Autonomous Water Quality Monitoring System

"Clean Water, Safe Future"

Team CircuitBotics · New Delhi, India

THE PROBLEM

Pollution Is Growing Faster Than We Can Monitor It



Chemicals & Farm Runoff

raise pH and contaminate lakes and ponds

Puts aquatic life and communities at risk



Slow, Expensive Testing

needs trained experts on-site, again and again

Not practical for daily checks, water body by water body



Damage Already Done

by the time pollution is finally detected

Fish die off and water turns unsafe for irrigation



5,000,000+

ponds & lakes across India

— most go unmonitored

OUR SOLUTION

Meet AquaBot

An autonomous boat that patrols lakes and ponds, continuously measuring water quality — and takes action the moment something's wrong.

WHAT IT MONITORS



pH Level



TDS



Turbidity



Temperature



Rainfall



Dissolved Oxygen

Powered by a dual-controller design — a Raspberry Pi Zero 2W for sensing & AI, an ESP32 for real-time navigation — with live cloud, local Node-RED and TFT dashboards.



“Clean Water, Safe Future”

Affordable, Mobile — and It Actually Takes Action



Fixed Buoys

- ⊗ Only monitors a single fixed spot
- ⊗ Often costs lakhs of rupees



Manual Boats

- ⊗ Needs a person to operate it
- ⊗ No automatic response to problems



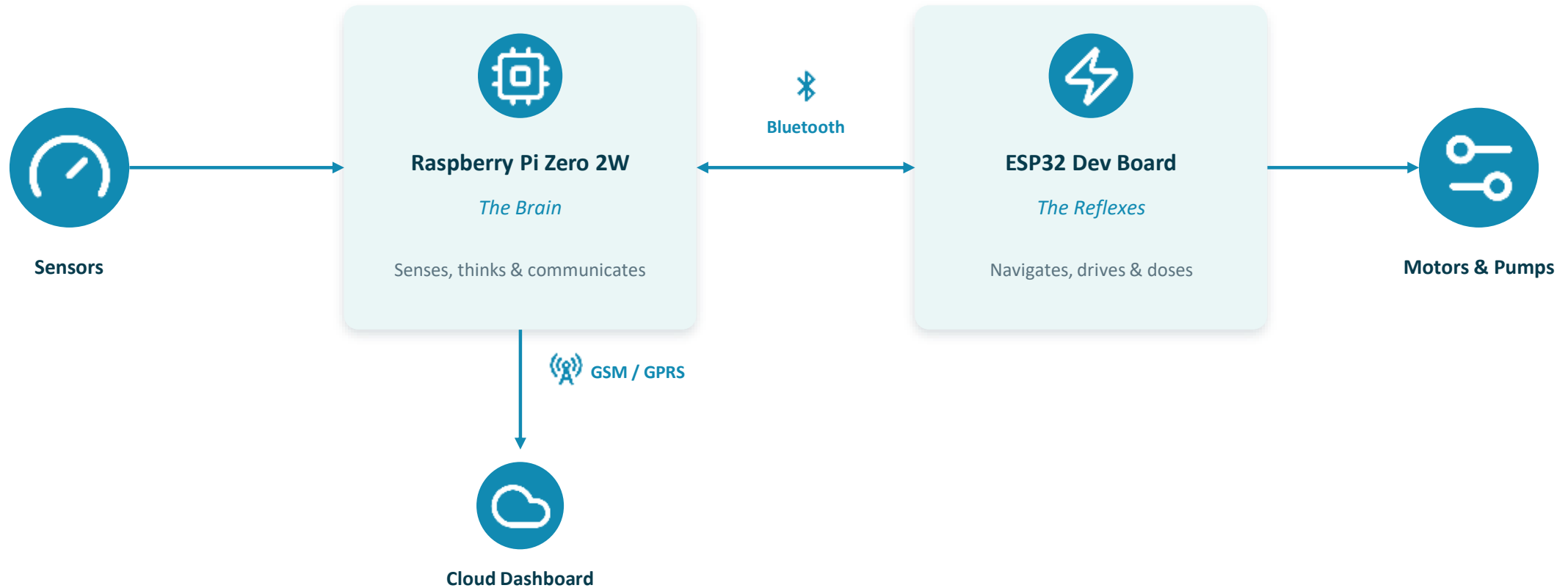
AquaBot

- ✓ Patrols autonomously, wide area
- ✓ Responds instantly, on its own

The right tool for continuous, wide-area protection.

Prototype cost: under ₹20,000 — versus lakhs of rupees for a commercial buoy

Two Brains, Working as One



Cloudflare Worker stores data in Workers KV; readings are mirrored locally via Node-RED.

FLEXIBLE CONTROL

Three Ways to Patrol



Manual

Operator drives live from the web dashboard — perfect for close inspections.

Best for: close inspections & live demos



Autonomous

Patrols a geo-fenced area on its own using GPS waypoints.

Best for: full-area surveys without constant attention



Stationary

Holds one exact position for long-duration spot monitoring.

Best for: drains & suspected pollution sources

Switch modes anytime — AquaBot always comes to a safe stop first.

It Doesn't Just Detect Problems — It Responds



Automatic pH Correction

- ✓ Detects unsafe pH in real time
- ✓ Sprays the exact corrective dose in seconds
- ✓ 30-second cooldown prevents overspraying

Safe range: pH 6.5–8.5, calibrated against reference solutions



AI Garbage Detection

- ✓ Camera + AI spot floating debris
- ✓ Captures an image, tags the GPS location
- ✓ Sends an instant alert to the cleanup crew

Runs on an Edge TPU — ~200ms per frame

Zero operator input required for either response.

ALWAYS-ON MONITORING

Live Data, Wherever You Are



Cloud Dashboard

Live data over GSM/GPRS — works with zero WiFi

Built on Cloudflare Workers + Workers KV



Node-RED (Local)

Backup dashboard on the boat's own network

Trimmed & memory-capped for the Pi Zero



Onboard TFT Screen

At-a-glance readout right on the boat

2.8-inch display — no internet needed



Live Camera Feed

See what the boat sees, when in WiFi range

Also feeds the AI garbage-detection model

Always reachable — on-site or across the country.

REAL-WORLD SCENARIO

From Detection to Action — in Minutes

A polluted lake, with no WiFi in sight

Industrial runoff has been raising the lake's pH for months — fish are dying and the community can no longer use the water for irrigation.

STEP 1



Deployed in Autonomous mode —
patrols the lake

STEP 2



Detects pH 9.2 — well above the safe
range

STEP 3



Auto-doses correction & alerts the
authority over GSM

STEP 4



AI spots floating plastic — geo-tags it
for cleanup

No WiFi. No delay. No damage.

Protecting Communities & Ecosystems



Local Communities

Depend on the lake every day



Farmers

Need safe water for irrigation



Fishermen

Livelihoods need healthy water



Municipal Authorities

Need affordable public monitoring

500–5,000 people per water body — even 1% deployment across India’s 5M+ ponds and lakes reaches millions.

SUPPORTING THE UN SUSTAINABLE DEVELOPMENT GOALS



Clean Water & Sanitation



Industry & Innovation



Life Below Water



Life on Land

OUR ROADMAP

From Prototype to Real-World Deployment

First 10 units estimated at ₹2–2.5 lakhs, with outreach to schemes like Jal Jeevan Mission and Startup India.



In five years: hundreds of lakes and ponds, guarded by AquaBot.

THE TEAM

Team CircuitBotics



Vivek Kumar

Team Leader

Hardware & software integration, navigation systems, dashboards



MD Arif

Team Member

Mechanical build, electronics assembly, research & presentation



Sanjay Bhatt

Coach

Guidance, mentorship & logistics throughout the project

Together, Vivek and Arif researched, built and programmed AquaBot from scratch.

DBRA CM Shri Janakpuri Desu Colony, New Delhi, India



“Every water body deserves a guardian.”

Thank You

Clean Water, Safe Future